

Domain expertise and scientific reading

An exploratory PoTeC analysis with lme4

Analysis produced with the lme4-mixed-models skill

2026-09-27

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Purpose and interpretation

This analysis asks whether reading one's own discipline rather than the other discipline is associated with different first-pass reading behavior, whether that contrast differs between undergraduate and graduate readers, and whether domain match modifies sensitivity to surprisal. Word length and lexical frequency are included as additional lexical predictors.

This is an **exploratory, informed trial of a statistical skill**. The analyst had seen the proposed evaluation criteria. There is no no-skill comparison and no basis for estimating the skill's causal benefit from this run. The scientific analysis itself is observational: discipline and study level were not randomized.

The continuous analysis uses 95,683 fixated-word observations (full word sample). The binary analysis uses 33,600 word opportunities (quarter word sample). Both preserve all 75 readers and 12 texts. The sampled-word fits use a seeded quarter of eligible word tokens within every text, shared across readers. They must not be described as full-data analyses.

Findings

The clearest pattern in this specification is a modest own-discipline reading-time advantage among graduate readers. At the reference lexical profile, the equally discipline-weighted own-minus-other contrast is **−0.0299 log milliseconds** (unadjusted 95% interval −0.0424 to −0.0174; twelve-test Holm-adjusted $p = 0.00034$). Exponentiating this log-scale contrast gives a geometric-mean ratio of about **0.971**, or **2.95% lower** first-pass reading time given fixation. This ratio is not an unqualified population arithmetic-mean effect.

For undergraduates, the corresponding estimate is −0.00785 log milliseconds (95% interval −0.0234 to 0.00773). The graduate-minus-undergraduate difference in domain-match contrasts is −0.0220 (95% interval −0.0427 to −0.00139), but its Holm-adjusted p is **0.379**. The result therefore does not establish that the domain-match

advantage differs by study level under the declared comparison family. One individually small p-value and one individually large p-value would not establish that difference either.

The graduate own-minus-other difference in surprisal slope is -0.0149 log milliseconds per surprisal SD (95% interval -0.0256 to -0.00423), with adjusted $p = \mathbf{0.073}$. The graduate-versus-undergraduate difference in this slope contrast has adjusted $p = 1$. These results do not establish study-level moderation of the expertise-surprisal association. They are also not equivalence tests.

None of the sampled-word FPreReg model’s six exploratory Wald contrasts has a small adjusted p-value. However, the joint derivative check timed out; these approximate inferences are **numerically unqualified**, so this is not a validated negative finding. Its outcome is regression per eligible word opportunity, not regression conditional on fixation. The response curves illustrate a substantial distinction between zero-random-effect and integrated probabilities, but their point differences should not be interpreted without the uncertainty and sampling limits.

Adequacy limits matter here. The log-FPRT residual Q–Q plot has heavier tails than a Gaussian distribution, and standardized residual SD varies from about 0.89 to 1.17 across fitted-value deciles. Adjacent-word residual correlation is small (0.0145 over 63,931 eligible pairs), but that alone does not validate the residual model. The GLMM’s conditional simulation checks place its observed adjacent residual correlation and reader–text rate spread inside the simulated 95% ranges; these limited in-sample checks do not establish overall adequacy.

All fitted covariance specifications were singular. The same-row correlated reader/item LMM sensitivity, evaluated on the quarter-word sample, changed the six focal point estimates by less than 0.19 baseline standard errors, but its Hessian is indefinite. That is limited evidence of point-estimate stability, not a clean robustness pass. It does not replace a full-data covariance sensitivity. The stricter full-data LMM changed the focal contrasts by less than 0.00001 standard errors. Additional numerical audit results are shown below.

The explicit GLMM derivative calculation reached its 900-second limit without returning a Hessian. The warm-start alternative LMM optimizer reached its 600-second limit without returning a fit. These are recorded incomplete checks, not successes. The GLMM’s reported exploratory standard errors use lme4’s RX/PIRLS approximation, not the unavailable joint Hessian. The full LMM’s strict refit supports point-estimate stability but retains some scaled-gradient concerns.

The appropriate conclusion is an **exploratory own-domain reading-time association among graduate readers, with unresolved covariance/adequacy qualifications**. This is not a causal effect of training, a validated population prediction, or a confirmatory result about how expertise changes surprisal processing. The LMM remains under SINGULAR_REVIEW with Gaussian adequacy concerns; full numerical qualification remains NUMERICAL_UNRESOLVED for the LMM, GLMM and covariance sensitivity. This run is not assigned READY_WITH_LIMITS.

This bounded trial did not complete a full-corpus GLMM, a GLMM covariance sensitivity, a fixation-conditioned FPreReg sensitivity, leave-text-out influence fits, or refit-based intervals propagating variance-component uncertainty. Those omissions limit the scientific claims made here.

Data provenance and the observation unit

The inputs come from the [PoTeC repository](#) and its linked OSF word-level reading-measures archive. Git revision 46247ca2aea5876311acf6b59338880d0bf5449d supplies the documentation and lexical features. The separately downloaded OSF archive has its own SHA-256 in `data-manifest.json`; the Git revision alone does not pin that archive. The repository records a surprisal correction on June 17, 2026. Every selected surprisal value in the archive was checked against the corrected pinned features.

The audit found **142,125 unique reader $\times$ text $\times$ word-position observations**, 75 readers and 1,895 word tokens. The roughly 404,000 count discussed in the benchmark proposal belongs to a different, scanpath-level table. Treating fixation-level repetitions as independent word outcomes would change the unit of analysis.

rows	missing frequency	skipped	first pass fixated	regressions
142125	8625	38408	103717	23065

reader discipline numeric	level of studies numeric	readers
1	1	20
1	0	12
0	1	27
0	0	16

The upstream merge script fills missing lexical features with zero. The pinned original word-feature files identify 115 words with unavailable lemma frequency, corresponding to 8,625 reader–word rows. These were restored to missing and excluded from this complete-covariate pilot. The remaining set has 133,500 binary-outcome rows and 95,683 first-pass-fixated rows. This restriction changes the word population; it is not evidence that lexical missingness is ignorable.

The analysis uses text-context GPT-2-large surprisal, chosen before fitting. Log frequency, word length and surprisal are centered and scaled using **unique eligible word tokens**, rather than the fixation-selected outcome rows.

variable	center	scale
W	5.881	3.544
F	5.972	4.105
S	4.373	3.695

Outcomes, design and estimands

For the LMM, the response is $\log(\text{FPRT})$ among $\text{FPF} == 1$ observations with strictly positive FPRT. The audit verified that positive FPRT agrees with FPF. Skipped words are not treated as zero-millisecond readings and no arbitrary constant is added before logging.

For the GLMM, FPre_g is modeled over **all eligible word opportunities**, including zeros for words not fixated in the first pass. The audit verified that unfixated words have FPre_g zero. This outcome combines fixation opportunity and regression propensity. It is different from the probability of regression conditional on first-pass fixation; the conditional analysis is a sensitivity target, not an interchangeable denominator.

reader discipline	study level	text domain	readers	words	regressions	non regressions
Physics	Graduate	Biology	20	4480	651	3829
Physics	Graduate	Physics	20	4480	812	3668
Physics	Undergraduate	Biology	12	2688	415	2273
Physics	Undergraduate	Physics	12	2688	474	2214
Biology	Graduate	Biology	27	6048	762	5286
Biology	Graduate	Physics	27	6048	1009	5039
Biology	Undergraduate	Biology	16	3584	488	3096
Biology	Undergraduate	Physics	16	3584	548	3036

Each of the eight factorial cells contains regressions and non-regressions. This rules out complete separation by those cells alone; it does not prove absence of separation along every continuous predictor combination.

Readers are crossed with word tokens. Tokens belong to texts. Multiple words also share a reader–text reading episode. Text domain and lexical covariates vary within readers; reader discipline and study level vary within word items. A by-word slope for word length or text domain would not represent within-word variation. The support audit is retained in `outputs/design_support.csv`.

Numeric contrasts are fixed before fitting: biology = −0.5 and physics = +0.5 for reader discipline D and text domain T; undergraduate = −0.5 and graduate = +0.5 for study level L. S, F, and W denote standardized surprisal, log lemma frequency and word length. TS=T*S and DL=D*L are explicit numeric random-effect columns.

For each study level, the domain-match estimand is

$$\Delta_l = \frac{1}{2}\{\eta_{B,l,B} - \eta_{B,l,P} + \eta_{P,l,P} - \eta_{P,l,B}\},$$

at the mean lexical profile. Reader disciplines receive equal weight, rather than their unequal sample proportions. We also estimate $\Delta_{graduate} - \Delta_{undergraduate}$, and the same three contrasts of the surprisal slope. These are linear-predictor contrasts: log milliseconds for the LMM and log odds for the GLMM. They are not probability differences.

Model specification and computation

The fixed structure is

```
D * L * T * S + (D * T) * (F + W)
```

This preserves the factorial hierarchy, includes study-level moderation of the expertise-surprisal relationship, and allows domain-match dependence of the length and frequency associations. Study-level moderation of length/frequency is outside this pilot’s fixed-effect specification.

The starting random structure is

```
(1 + T + S + TS + W + F || reader) +  
(1 + D + L + DL || item) +  
(1 + D + L + DL || text) +  
(1 | reader_text)
```

All double-bar predictors are numeric. This deliberately diagonal covariance structure is a pragmatic restriction in the recorded contrast/centering basis. It is not invariant to arbitrary recoding and is not claimed to be a maximal model. Reader domain and domain-by-surprisal slopes protect the central within-reader questions; item and text slopes represent reader-group variation within stimuli. Only 12 texts inform the text covariance components.

The LMM uses REML. The GLMM uses a binomial logit likelihood with Laplace approximation (`nAGQ=1`). Gradient and Hessian calculation was explicitly enabled; installed lme4 defaults can omit derivative checks for large models. No fixed-effect selection or significance-driven random-effect deletion was used. The analysis plan and bounded computational policy are in `analysis-plan.yml`.

model	observations	elapsed seconds	singular	optimizer code	derivatives available
Log-FPRT LMM	95683	560.367	TRUE	0	TRUE
FPReg GLMM	33600	793.403	TRUE	0	FALSE

Optimizer exit codes are not adequacy certificates. The retained audits include warnings, `VarCorr`, `rePCA`, gradients and Hessian eigenvalues. Boundary estimates were retained and examined rather than removed until a warning disappeared.

Installed lme4 2.0.1 did not provide the singular GLMM’s derivatives despite `calc.derivs=TRUE`. The extra check reconstructs the same deviance function, verifies its objective against the fitted likelihood, and explicitly evaluates finite-difference derivatives. The original fit and original audit are preserved.

Explicit derivative check timed out after 900 seconds; no Hessian result is available. Numerical qualification remains unresolved.

Table 6: Log-FPRT LMM variance components

grp	var1	var2	vcov	sdcov
item	DL	NA	0.000	0.001
item.1	L	NA	0.000	0.000
item.2	D	NA	0.004	0.064
item.3	(Intercept)	NA	0.034	0.184
reader_text	(Intercept)	NA	0.002	0.039
reader	F	NA	0.001	0.031

Table 6: Log-FPRT LMM variance components (*continued*)

grp	var1	var2	vcov	sdcor
reader.1	W	NA	0.002	0.040
reader.2	TS	NA	0.000	0.021
reader.3	S	NA	0.001	0.022
reader.4	T	NA	0.000	0.019
reader.5	(Intercept)	NA	0.016	0.126
text	DL	NA	0.000	0.008
text.1	L	NA	0.000	0.000
text.2	D	NA	0.000	0.000
text.3	(Intercept)	NA	0.000	0.015
Residual	NA	NA	0.198	0.445

Table 7: FPReg GLMM variance components

grp	var1	var2	vcov	sdcor
reader_text	(Intercept)	NA	0.000	0.000
item	DL	NA	0.012	0.110
item.1	L	NA	0.000	0.000
item.2	D	NA	0.032	0.178
item.3	(Intercept)	NA	0.736	0.858
reader	F	NA	0.008	0.087
reader.1	W	NA	0.010	0.100
reader.2	TS	NA	0.000	0.000
reader.3	S	NA	0.000	0.000
reader.4	T	NA	0.008	0.089
reader.5	(Intercept)	NA	0.193	0.440
text	DL	NA	0.032	0.180
text.1	L	NA	0.012	0.110
text.2	D	NA	0.000	0.000
text.3	(Intercept)	NA	0.026	0.161

The full coefficient tables below describe the recorded contrast basis. They are not substitutes for the planned domain-match contrasts.

Table 8: Log-FPRT LMM fixed coefficients

term	estimate	SE
(Intercept)	5.5147	0.0165
D	0.0532	0.0306
L	0.0205	0.0306
T	0.0022	0.0134
S	0.0642	0.0062
F	-0.0090	0.0085
W	0.1580	0.0088
D:L	-0.0434	0.0612
D:T	-0.0377	0.0100
L:T	-0.0060	0.0095
D:S	-0.0011	0.0067
L:S	-0.0092	0.0064
T:S	-0.0195	0.0115
D:F	0.0014	0.0090
D:W	0.0026	0.0106
T:F	0.0223	0.0154
T:W	0.0110	0.0148
D:L:T	-0.0441	0.0195
D:L:S	-0.0017	0.0127
D:T:S	-0.0178	0.0093
L:T:S	-0.0067	0.0078
D:T:F	0.0380	0.0108

Table 8: Log-FPRT LMM fixed coefficients (*continued*)

term	estimate	SE
D:T:W	-0.0094	0.0101
D:L:T:S	-0.0240	0.0155

Table 9: FPRreg GLMM fixed coefficients

term	estimate	SE
(Intercept)	-2.0669	0.0840
D	0.1846	0.1123
L	-0.0209	0.1161
T	0.2534	0.1303
S	0.0536	0.0561
F	-0.1885	0.0771
W	0.1952	0.0773
D:L	-0.0275	0.2295
D:T	-0.0269	0.0752
L:T	0.1664	0.0955
D:S	0.0104	0.0413
L:S	0.0260	0.0369
T:S	0.0428	0.1122
D:F	0.0355	0.0592
D:W	0.0191	0.0597
T:F	0.3198	0.1528
T:W	0.2102	0.1527
D:L:T	-0.0780	0.1770
D:L:S	0.1665	0.0740
D:T:S	0.0700	0.0825
L:T:S	0.0243	0.0698
D:T:F	0.1768	0.1109
D:T:W	0.1254	0.1094
D:L:T:S	0.1470	0.1400

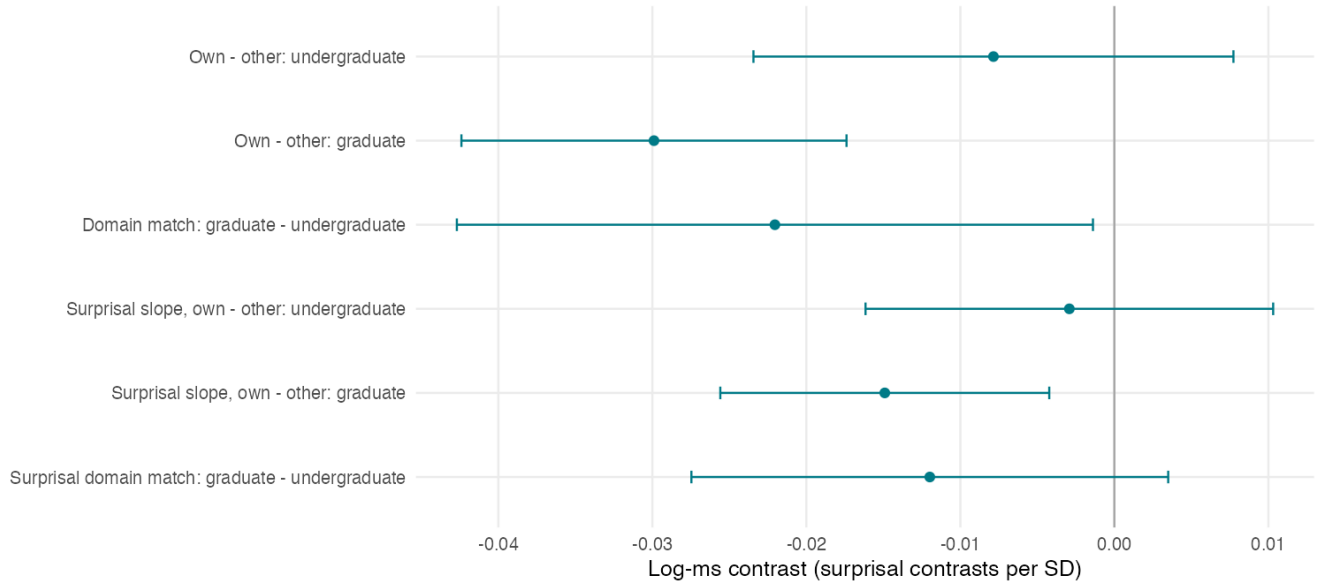
Planned contrasts and inference

The LMM uses Satterthwaite approximate t inference through `lmerTest`; the GLMM uses exploratory asymptotic Wald z inference with lme4’s RX/PIRLS covariance approximation. The explicit GLMM joint-Hessian calculation timed out, so these GLMM intervals and p-values are numerically unqualified and are retained for transparency, not as validated scientific inference. Satterthwaite/Kenward–Roger corrections are not applied to the logistic model. The twelve planned tests—six per outcome—form one Holm-adjusted family. The displayed 95% intervals are **unadjusted**. These are model-based approximations, not simulation-calibrated guarantees.

outcome	contrast	estimate	95% interval	Holm p
lmm	Own - other: undergraduate	-0.0079	[-0.0234, 0.0077]	1.0000
lmm	Own - other: graduate	-0.0299	[-0.0424, -0.0174]	0.0003
lmm	Domain match: graduate - undergraduate	-0.0220	[-0.0427, -0.0014]	0.3792
lmm	Surprisal slope, own - other: undergraduate	-0.0029	[-0.0162, 0.0103]	1.0000
lmm	Surprisal slope, own - other: graduate	-0.0149	[-0.0256, -0.0042]	0.0729
lmm	Surprisal domain match: graduate - undergraduate	-0.0120	[-0.0275, 0.0035]	1.0000
glmm	Own - other: undergraduate	0.0060	[-0.1177, 0.1297]	1.0000
glmm	Own - other: graduate	-0.0329	[-0.1360, 0.0701]	1.0000
glmm	Domain match: graduate - undergraduate	-0.0390	[-0.2125, 0.1345]	1.0000
glmm	Surprisal slope, own - other: undergraduate	-0.0018	[-0.1186, 0.1151]	1.0000
glmm	Surprisal slope, own - other: graduate	0.0717	[-0.0222, 0.1656]	1.0000
glmm	Surprisal domain match: graduate - undergraduate	0.0735	[-0.0637, 0.2107]	1.0000

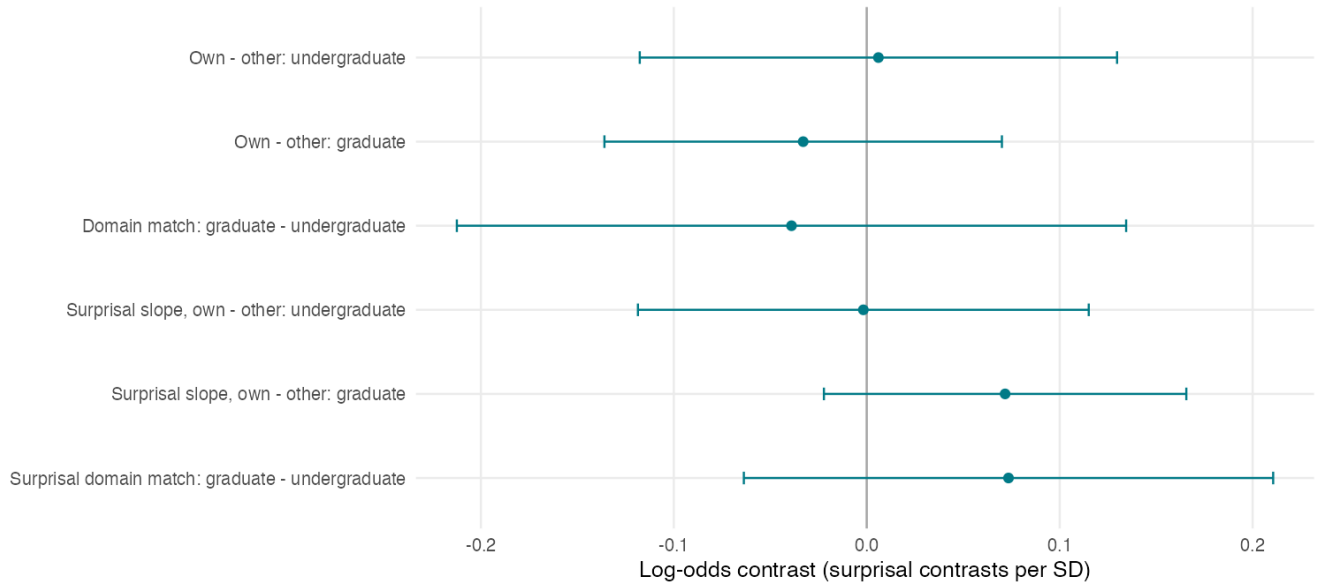
Planned domain-match contrasts

Unadjusted 95% intervals; 12-test Holm-adjusted p-values in the table



Planned domain-match contrasts

Unadjusted 95% intervals; 12-test Holm-adjusted p-values in the table



Response scales and observed data

For new-group predictions, the variance of the random linear predictor is $v = zGz^\top$, summed across the fitted reader, item, text and reading-episode blocks. A lognormal arithmetic mean is $\exp(x\beta + (v + \sigma^2)/2)$. Simply exponentiating the log-scale expectation does not produce that mean.

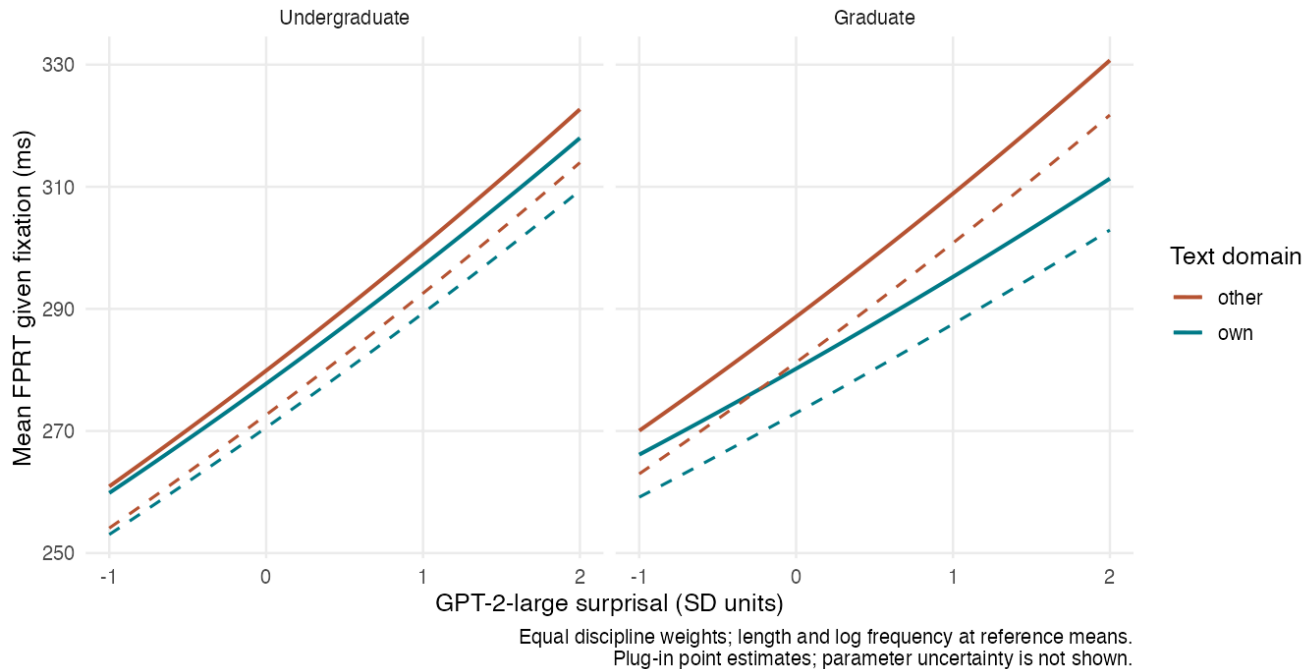
For the GLMM, the new-group probability integrates $\text{logit}^{-1}(x\beta + u)$ over $u \sim N(0, v)$. Gauss-Hermite quadrature was checked using two quadrature sizes and independent adaptive numerical integration. Setting random effects to zero instead yields $\text{logit}^{-1}(x\beta)$, a distinct target.

The curves below show **plug-in point estimates**, with estimated variance components held fixed. They have no uncertainty ribbons and should not be read as precise population curves. The contrast plots above display

fixed-effect uncertainty. A full interval for the integrated response mean would also need to propagate covariance-parameter uncertainty. Solid curves integrate over new-group heterogeneity; dashed curves set group effects to zero. For the LMM, both arithmetic-mean curves include residual lognormal retransformation.

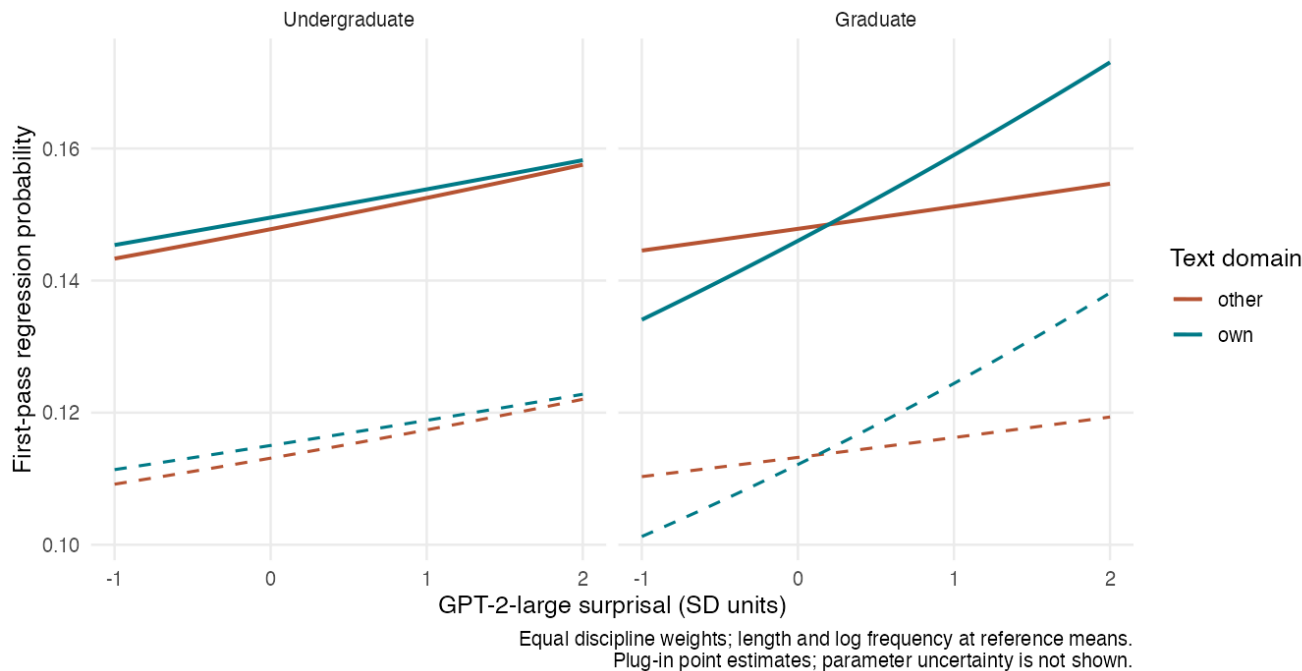
Reading time and surprisal

Solid: integrated new-group mean.
Dashed: random effects set to zero.



First-pass regressions and surprisal

Solid: integrated new-group mean.
Dashed: random effects set to zero.

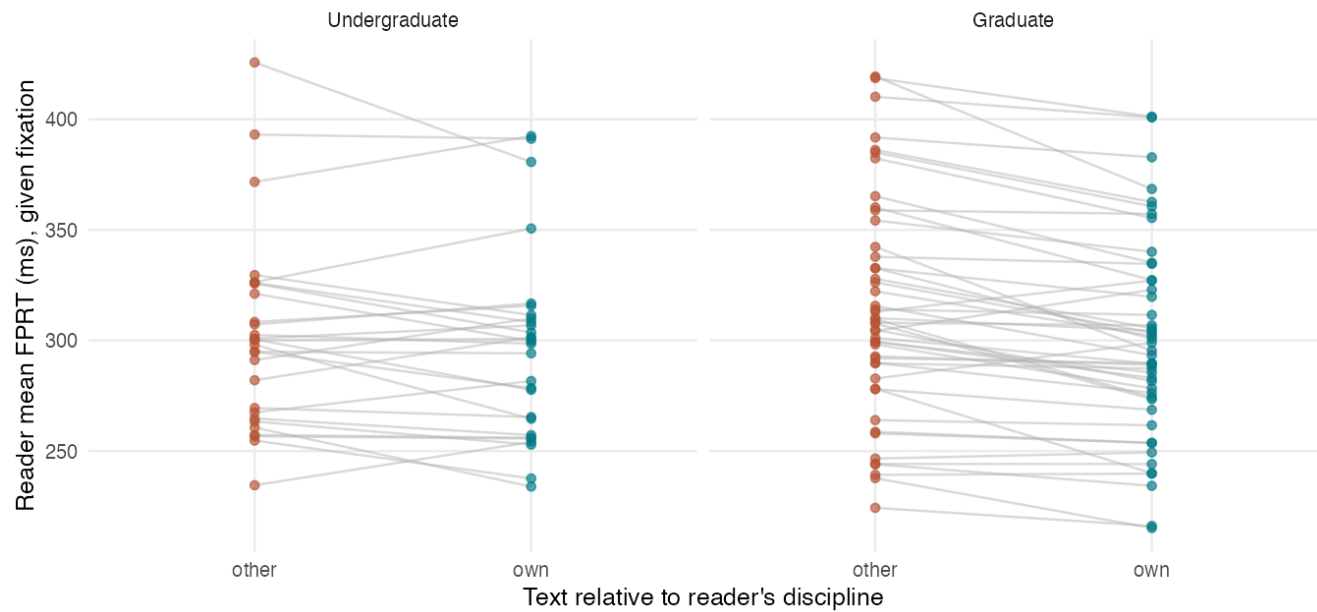


outcome	level	match	zero RE	integrated
lmm	Undergraduate	own	270.5590	277.7482
lmm	Undergraduate	other	272.6386	279.8830
lmm	Graduate	own	272.9473	280.1999
lmm	Graduate	other	281.2337	288.7065
glmm	Undergraduate	own	0.1150	0.1495
glmm	Undergraduate	other	0.1131	0.1478
glmm	Graduate	own	0.1122	0.1460
glmm	Graduate	other	0.1132	0.1478

Observed reader summaries provide context without treating thousands of words as independent reader replicates. These raw means mix lexical profiles and are not the same estimands as the adjusted reference-profile predictions.

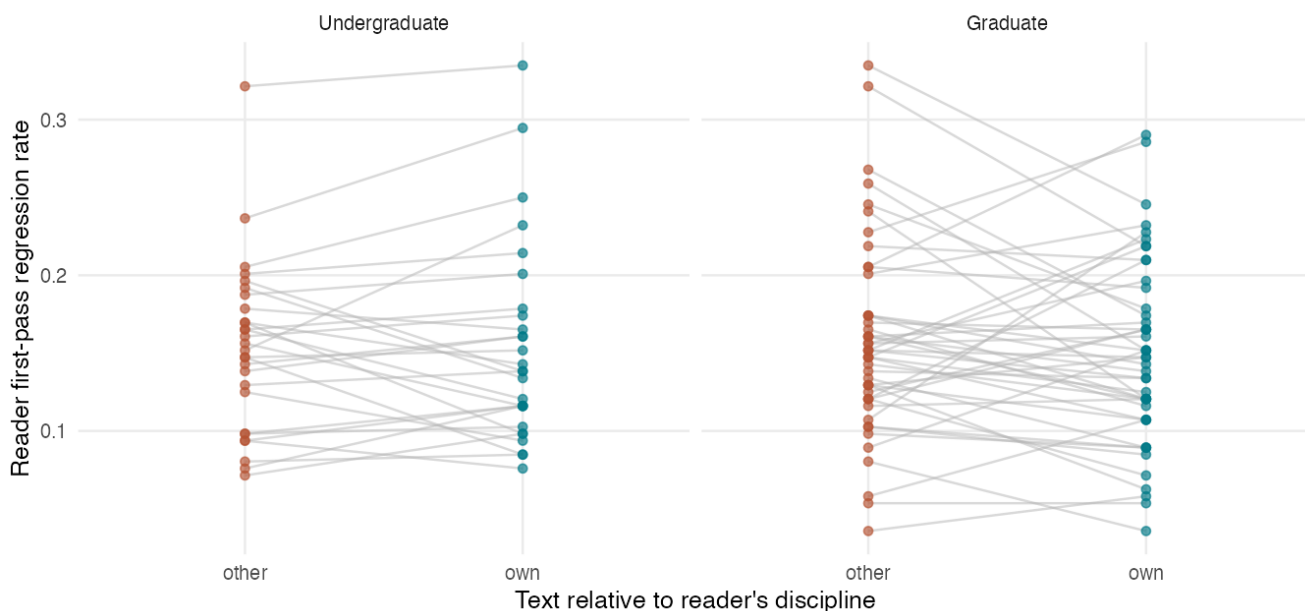
Observed reader summaries

Each point summarizes one reader and domain-match condition; denominators retained in CSV



Observed reader summaries

Each point summarizes one reader and domain-match condition; denominators retained in CSV



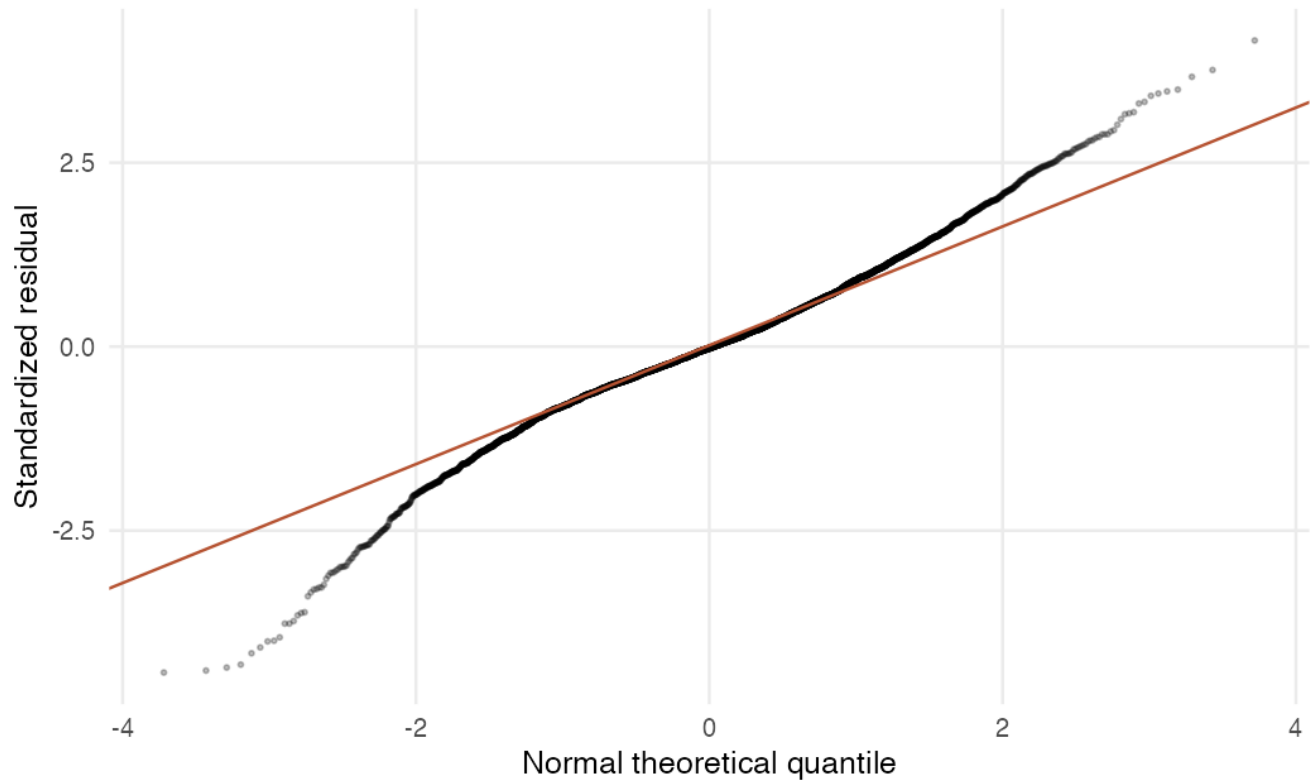
Adequacy and sensitivity

outcome	adjacent pairs	adjacent residual correlation	residual sd min	residual sd max	min conditional prediction	max conditional prediction
lmm	63931	0.0145	0.8873	1.1710	4.6323	7.9143
glmm	6675	-0.0223	0.6214	1.0276	0.0066	0.8435

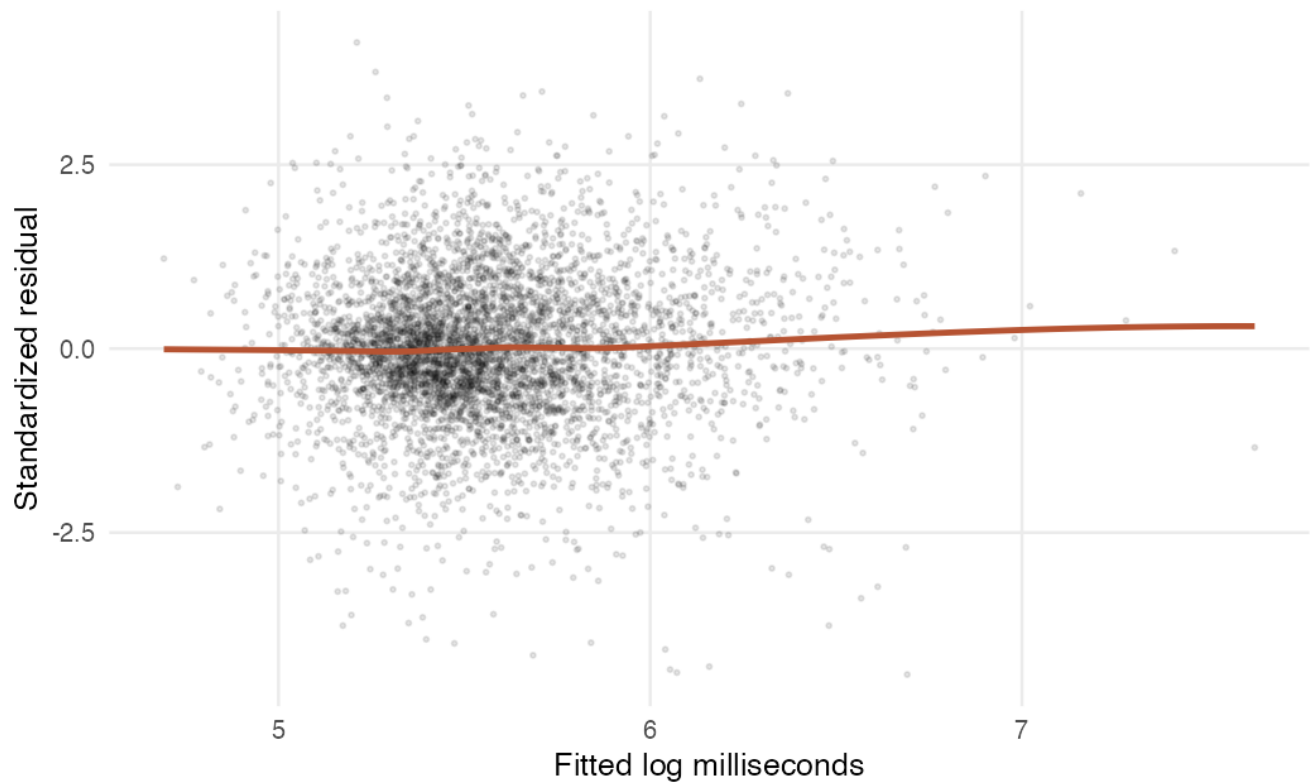
statistic	observed	simulated lower	simulated upper	nsim	mode
adjacent	-0.0223	-0.0239	0.0232	200	Conditional on beta and all fitted random effects; no refitting
group_rate_sd	0.0821	0.0774	0.0838	200	Conditional on beta and all fitted random effects; no refitting

Adjacent-word residual pairs are formed only within a reader–text episode and only where original word positions differ by one. Gaps introduced by subsampling or missing covariates are not treated as adjacent words. Binary simulation checks use 200 Bernoulli simulations conditional on fitted fixed and random effects, without refitting. They probe residual structure at those fitted values; they do not validate generalization or calibrate inferential error rates. DHARMA was not installed, so these explicit checks are not labelled DHARMA results.

LMM conditional residual Q-Q plot

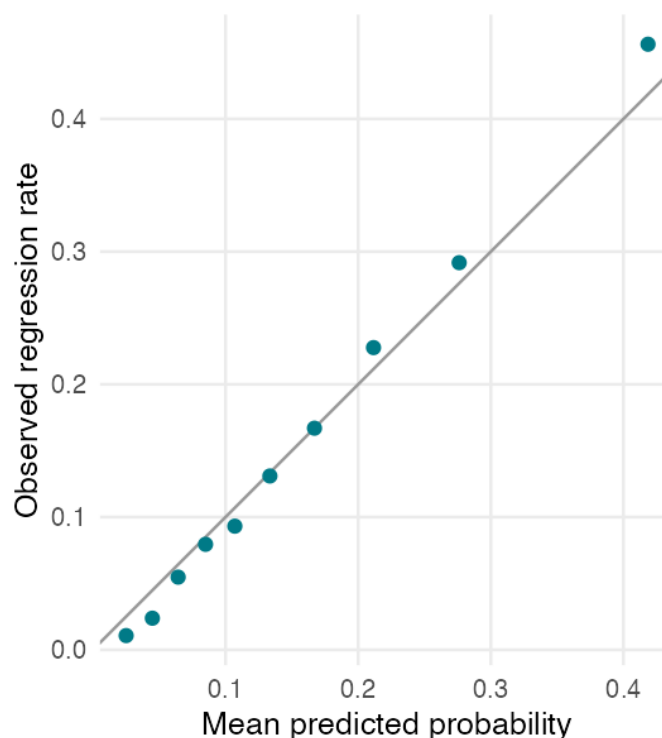


LMM conditional residuals



GLMM conditional calibration

In-sample deciles; descriptive, not held-out validation



model	contrast	estimate	Shift / baseline SE
lmm-full-diagonal-strict	Own - other: undergraduate	-0.0079	0.0000
lmm-full-diagonal-strict	Own - other: graduate	-0.0299	0.0000
lmm-full-diagonal-strict	Domain match: graduate - undergraduate	-0.0220	0.0000
lmm-full-diagonal-strict	Surprisal slope, own - other: undergraduate	-0.0029	0.0000
lmm-full-diagonal-strict	Surprisal slope, own - other: graduate	-0.0149	0.0000
lmm-full-diagonal-strict	Surprisal domain match: graduate - undergraduate	-0.0120	0.0000
lmm-quarter-correlated-primary	Own - other: undergraduate	-0.0039	0.1258
lmm-quarter-correlated-primary	Own - other: graduate	-0.0362	-0.1197
lmm-quarter-correlated-primary	Domain match: graduate - undergraduate	-0.0323	-0.1806
lmm-quarter-correlated-primary	Surprisal slope, own - other: undergraduate	0.0035	-0.0355
lmm-quarter-correlated-primary	Surprisal slope, own - other: graduate	-0.0065	-0.0241
lmm-quarter-correlated-primary	Surprisal domain match: graduate - undergraduate	-0.0100	0.0139

Sensitivity fits are judged by focal estimate changes and numerical evidence, not by which produces smaller p-values. Correlated reader/item covariance is the prespecified structural alternative. A fixation-conditioned FPreReg model, if completed, changes the outcome population and must be interpreted accordingly. Remaining sensitivity and adequacy limitations are stated in the findings.

Reproduction and retained evidence

The document executes R to inspect retained fits and produce tables; expensive fitting is separated into scripts so rendering does not silently launch hours of refitting. The scripts and fitting formulas remain editable and reproducible:

```
python3 fetch_data.py
Rscript --vanilla prepare.R
Rscript --vanilla fit.R lmm full diagonal
Rscript --vanilla fit.R glmm quarter diagonal
Rscript --vanilla summarize.R full quarter
Rscript --vanilla sensitivity.R
quarto render analysis.qmd
```

Use the actual fit scopes and additional commands listed in `findings.md`. Existing downloaded inputs are retained; verify `data-manifest.json` before reuse. Re-running with a new download may produce different upstream bytes. The prose findings describe this frozen run; substantive refits require reviewing that interpretation as well as regenerating numerical tables. The raw log and JSON receipt for each controlled run retain its exit status, timing and command. No source skill or pre-existing repo edits were changed. The CSV contrast table also retains standard errors, denominator degrees of freedom, unadjusted p-values, interval limits and inference methods.

```
R version 4.5.1 (2025-06-13)
Platform: aarch64-apple-darwin20
Running under: macOS Sonoma 14.3

Matrix products: default
BLAS: /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework/Versions/A/libBLAS.dylib
LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib; LAPACK version 3.12.1

locale:
[1] C

time zone: America/Toronto
tzcode source: internal

attached base packages:
[1] stats      graphics  grDevices  utils      datasets  methods   base

other attached packages:
[1] knitr_1.51      ggplot2_4.0.3   lme4_2.0-1      Matrix_1.7-3
[5] data.table_1.18.4

loaded via a namespace (and not attached):
[1] gtable_0.3.6      jsonlite_2.0.0    dplyr_1.2.1
[4] compiler_4.5.1    tidyselect_1.2.1  Rcpp_1.1.2
[7] xml2_1.6.0         stringr_1.6.0     textshaping_1.0.5
[10] systemfonts_1.3.2 splines_4.5.1     scales_1.4.0
[13] boot_1.3-31        yaml_2.3.12       fastmap_1.2.0
[16] lattice_0.22-7     R6_2.6.1          generics_0.1.4
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[22] tibble_3.3.1       nloptr_2.2.1      svglite_2.2.2
[25] minqa_1.2.8        pillar_1.11.1     RColorBrewer_1.1-3
[28] rlang_1.3.0         stringi_1.8.9     xfun_0.60
[31] S7_0.2.2           otel_0.2.0        viridisLite_0.4.3
[34] cli_3.6.6          withr_3.0.3       magrittr_2.0.5
[37] Rdpack_2.6.6       digest_0.6.39     grid_4.5.1
[40] rstudioapi_0.18.0  lifecycle_1.0.5   nlme_3.1-170
[43] reformulas_0.4.4   vctr_0.7.3        evaluate_1.0.5
[46] glue_1.8.1         farver_2.1.2      rmarkdown_2.31
[49] pkgconfig_2.0.3    tools_4.5.1       htmltools_0.5.9
```

What this trial showed about the skill

The skill successfully prompted an explicit analysis contract before fitting, an audit of the observation unit, design-supported reader/item slopes, recorded contrast coding, separation of singularity from convergence, and explicit prediction targets. The data work found a consequential mismatch between the proposed observation count and the word-level table, and restored lexical missingness that the upstream merge had replaced by zeros.

Those are observed properties of this analyst's run. They do not show that the same model without the skill would have failed, and some were already highlighted in the preceding conversation. This is not a measured skill improvement score.

The clearest weakness exposed by execution is the **numerical audit's incomplete visibility into skipped checks**. With installed lme4 2.0.1, a singular GLMM fit returned no derivative matrix even with `calc.derivs=TRUE`; a singular fit also bypasses the usual convergence-check path. The skill's helper reported singularity but did not specifically flag absent derivatives. A separate explicit derivative audit was needed; it exceeded its 900-second

limit and did not complete. The correlated LMM sensitivity fit returned an indefinite Hessian despite an optimizer success code. These are concrete reasons to improve the audit's evidence fields and stopping rules.

The trial also exposed a portability issue in the analysis code: converting a saved `lmerMod` to `lmerModLmerTest` initially failed because the recorded call referenced local data/formula/control objects. Rebinding the exact saved model frame corrected the reconstruction. The original failure log is retained.

The workflow was computationally expensive: a quarter-word GLMM required about 13 minutes, and the full LMM about 9 minutes, before all post-fit checks. A future skill revision would benefit from a staged runtime plan that preserves fitted objects before expensive checks, and explicit rules for reporting incomplete sensitivities. No change was made to the source skill during this trial.